FLAJOLET MARTIN [FM] ALGORITHM

Counting Distinct Elements

- Problem: a data stream consists of elements chosen from a set of size n. Maintain a count of the number of distinct elements seen so far.
- Obvious approach: maintain the set of elements seen.

Applications

- How many different words are found among the Web pages being crawled at a site?
  - Unusually low or high numbers could indicate artificial pages (spam?).
- How many unique users visited Facebook this month?
- How many different pages link to each of the pages we have crawled.

Flajolet-Martin Approach

- Pick a hash function h that maps each of the n elements to at least $\log_2 n$ bits.
- For each stream element a, let $r(a)$ be the number of trailing 0's in $h(a)$.
- Record R = the maximum $r(a)$ seen.
- Estimate = 2^R.

Flajolet Martin Algorithm

- **F**lajolet-Martin algorithm approximates the number of unique objects in a stream or a database in one pass.
- **I**f the stream contains n elements with m of them unique, this algorithm runs in O(n) time and needs O(log(m)) memory.

FM

- The Flajolet–Martin algorithm is an algorithm for approximating the number of distinct elements in a stream with a single pass.
- Space-consumption logarithmic in the maximal number of possible distinct elements in the stream.

EXAMPLE

Determine the distinct element in the stream using FM.

- Input stream of integers x = 1,3,2,1,2,3,4,3,1,2,3,1 .

EXAMPLE

Determine the distinct element in the stream using FM.

- **I**nput stream of integers x = 1,3,2,1,2,3,4,3,1,2,3,1 .
- **H**ash Function, $h(x) = 6x + 1 \bmod 5$

Calculate Hash Function h(x)

- For the given input stream 1,3,2,1,2,3,4,3,1,2,3,1
- $h(x) = 6x + 1 \mod 5.$

$$
\begin{aligned}
h(1) &= 6\,(1) + 1 \mod 5 \\
&= 6+1 \mod 5 \\
&= 7 \mod 5 \\
&= 2
\end{aligned}
$$

Therefore $h(1) = 2$

Similarly, calculate hash function for the complete stream.

CALCULATE HASH FUNCTION h(x)

- For the given input stream 1,3,2,1,2,3,4,3,1,2,3,1.

  $h(x) = 6x + 1 \mod 5$

$h(1) = 2$	$h(4) = 0$
$h(3) = 4$	$h(3) = 4$
$h(2) = 3$	$h(1) = 2$
$h(1) = 2$	$h(2) = 3$
$h(2) = 3$	$h(3) = 3$
$h(3) = 4$	$h(1) = 2$

BINARY BITS

- For the given input stream 1,3,2,1,2,3,4,3,1,2,3,1.
- For every hash function calculated, write the binary equivalent for the same.

h(1) = 2 = 010	h(4) = 0 = 000
h(3) = 4 = 100	h(3) = 4 = 100
h(2) = 3 = 011	h(1) = 2 = 010
h(1) = 2 = 010	h(2) = 3 = 011
h(2) = 3 = 011	h(3) = 4 = 100
h(3) = 4 = 100	h(1) = 2 = 010

TRAILING ZEROS

- For the given input stream 1,3,2,1,2,3,4,3,1,2,3,1.
- For every hash function calculated and the binary equivalent for the same is written.
- Now write the count of trailing zeros in each hash function bit.

h(1) = 2 = 010 = 1	h(4) = 0 = 000 = 0
h(3) = 4 = 100 = 2	h(3) = 4 = 100 = 2
h(2) = 3 = 011 = 0	h(1) = 2 = 010 = 1
h(1) = 2 = 010 = 1	h(2) = 3 = 011 = 0
h(2) = 3 = 011 = 0	h(3) = 4 = 100 = 2
h(3) = 4 = 100 = 2	h(1) = 2 = 010 = 1

DISTINCT ELEMENTS

- From the binary equivalent trailing zero values, write the value of maximum number of trailing zeros.
- The value of $r = 2$
- The distinct value $R = 2^r$
- Therefore $R = 2^2 = 4$
- Hence , there are 4 distinct elements as 1,3,2,4

Thank you